

# Matrices / Determinates

$|A| = 0 \infty$  Consistent

$|A| \neq 0$  Unique

Inconsistent

## i. Rank

→  $\det(A) \neq 0$  rank is  $n$  for  $n \times n$  Matrix

→  $\det(A) = 0$

Check

Largest Non zero Minor Matrix

• At least one  $2 \times 2$  Matrix is zero rank = 1

• If all zero  $2 \times 2$  det = 0  
rank = 0

→ Rank = 0 Element becomes zero after row operation

→ No. of Non zero Matrix  $\Rightarrow$  Rank

## Types of Matrices

1. Singular  $\det(A) = 0$

2. Non singular  $\det(A) \neq 0$

3. Orthogonal  $AA^T = I$

4. Idempotent  $A^2 = A$

5. Nilpotent  $A^k = 0$

6. Involutory  $A^2 = I$   
 $A = A^{-1}$

7. Symmetric  $A^T = A$

8. Skew Symmetric  $A^T = -A$

Zero

Row

Column

Square

Diagonal

Scalar

Unit

Lower &

Upper triangular  
Matrix

Singleton (1 element)